Web Mining

Quang Huy Duong, Ph.D.

SoICT, HUST

Hanoi, 2025

Course Content (15 weeks)

1. Information Retrieval
2. Link Analysis
3. Opinion Mining
4. Information extraction
5. Recommendation Systems
6. Project Presentation

Capstone Project

- Group of 3-4 students $\approx$ 12 groups
- Proposal: week 3; Present week 9 & 10
- Topic:
  - Traffic/Demand Prediction
  - Reinforcement Learning in Data Mining
  - Information Retrieval
  - Opinion Mining
  - Recommendation systems
- Approache:
  1. Academic: Problem definition, challenges, proposed method(s), experiments on benchmarks + code, conclusion
  2. Practical system or Apps:

Academic Project Example

- Step 1: Select a problem, e.g., sentiment classification
  - Problem should be well defined
  - What is the input? E.g., sentence
  - What is the output? E.g., one of the 2 classes: positive/negative
- Step 2: Select a set of method(s)
  - E.g., SVM, LLM APIs
  - Motivation for the selected methods
  - Explain the methods: Input representation, training, prediction
- Step 3: Select benchmark datasets
  - E.g., IMDB Dataset of 50K Movie Reviews
  - Make sure it is available !!!
- Step 4: Implementation
  - Which language? Which libs?
  - If based on existing code, what is your modification?
- Step 5: Performance report and result discussion
  - What metrics are used? Why?
  - Don't forget to compare to baseline models
  - Strong and weak points of the methods

Evaluation

- Report: Latex (Overleaf), max $\approx$ 30 pages
- Code: Git (Github), Google Colab
- Strictly follow the schedule
- Teamwork
- Containing advanced knowledge (E.g., Deep Learning)